

A Flexure-Based Parallel Actuation Dual-Stage System for Large-Stroke Nanopositioning

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Abstract—This paper presents a novel parallel actuation dual-stage system that delivers nanometric positioning over a large displacement. Unlike those traditional dual-stage designs, the translator of fine actuator in the proposed design is mechanically connected to the coarse translator via the flexure mechanism, while the actuation coils of both the coarse and fine actuators lay underneath the translators in parallel. The merits of the proposed parallel actuation dual-stage design are mainly twofold. First, both the coarse and fine actuators utilize the moving-magnet configuration, hence the translators do not need to carry any cables for power supply. Second, the coarse motion can exhibit better dynamics and energy efficiency due to the minimized moving size and weight. In this work, an analytical current-force model is established for the coarse actuator considering higher order harmonic magnetic field, and based on the proposed model, the force ripple of coarse actuator is quantitatively analyzed both in theory and in practical. Furthermore, a disturbance observer is employed in the dual-feedback configuration to deal with the uncertainties with the proved asymptotic stability. The experimental results show that the proposed dual-stage positioning system is capable to achieve 20 nm step resolution with a root mean square error of 13.15 nm, and the 5 mm point-to-point positioning error can achieve less than 40 nm.

Index Terms—Disturbance observer, dual-stage system, flexure mechanism, force ripple, Halbach PM array, nanopositioning, parallel actuation, sliding mode control.

I. INTRODUCTION

POSITIONING system with micro/nanometric precision is an essential technology for a wide range of industries

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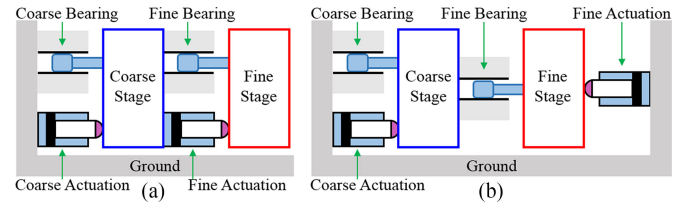


Fig. 1. (a) Series actuation and (b) parallel actuation dual-stage positioning system.

and research, such as atomic force microscopy, semiconductor manufacturing, and biocell manipulation. Normally, such high-performance positioning systems need to satisfy several requirements concurrently, such as large stroke, high accuracy, and wide servo bandwidth. As a result, it is hard for single-stage actuators to meet such requirements concurrently, e.g., the dynamic accuracy of linear motors are limited by the frictions, and the direct-drive voice coil motors (VCMs) are restricted in their motion stroke. Consequently, dual-stage configurations are adopted in many positioning systems, where the coarse actuators deliver large-displacement coarse motion with low precision and slow response, and the fine actuators provide fine motion with fast response and high accuracy [1]. The dual-stage positioning is widely applied in the hard disk drive industry [2], [3], where a small Pb–Zr–Ti (PZT) actuator is appended on a VCM. Similarly, one degree-of-freedom (DOF) dual-stage positioning systems are implemented in [4]–[6], and the dual-stage configurations are also applied in multi-DOF positioning systems [7], [8].

Traditionally, the actuation of dual-stage positioning systems are typically realized in series as shown in Fig. 1(a), where the fine actuator is fully mounted on the moving part of a coarse actuator. As a result, this design is more suitable for some compact PZT-based fine actuators [9], [10]. However, due to the well-known physical limitations of PZT actuators, e.g., small motion stroke and nonlinear effects [11], [12], electromagnetic actuators are also widely adopted for fine positioning. As the electromagnetic fine actuators contain actuation coils with certain volume and mass, such series actuation dual-stage structure leads to a noncompact design with relatively larger moving mass in the coarse translator, so that the coarse actuator dynamics is degraded unavoidably and energy consumption is also increased unnecessarily. Furthermore, the series actuation dual-stage design is unavoidable to have the undesired energy chains linked with the translator to power the fine actuators, which causes substantial disturbance and increases both the system complexity and cost.

In this paper, a novel dual-stage positioning system with parallel actuation is proposed as illustrated in Fig. 1(b). Differing from the traditional series actuation dual-stage design, only the fine translator is connected with the coarse translator in the proposed design, while the actuation coils of both the coarse and fine actuators lay underneath their translators accordingly. As a result, both the size and weight of the coarse translator can be minimized, which improve the dynamics performance and energy efficiency. Moreover, this design also provides a potential for a cable-free translator design in both coarse and fine actuators, since no energy chain attached with the coarse translator is necessary to power the fine actuator. The proposed dual-stage positioning system employs a Halbach permanent magnet (PM) array-based moving magnet linear motor (Halbach MMLM) as the coarse actuator and its fine actuator is a flexure-based Lorentz motor (FLM) [13]. The Halbach MMLM adopted in this system was first proposed in [14] and was used as the two DOF actuators in the six DOF magnetically levitated planar positioners [15]–[18]. The flexure mechanism is employed as the mechanical guide in the FLM to offer a motion bearing with no dry friction and backlash [19], [20]. Combined with the electromagnetic actuation, the FLM owns the advantages such as good linearity, fast response, relatively large motion range, and low cost.

Disturbances and model uncertainties are critical for the high-precision motion control, and observers are utilized as a solution to estimate the disturbance online and compensate based on the estimation [21]–[23]. On the other hand, the sliding mode control is also used to stabilize the system in the presence of disturbance and perturbation [24]–[26]. Furthermore, adaptive and robust controllers are also commonly used to deal with the uncertainties and disturbances, a backstepping method synthesizing the adaptive and robust control is proposed in [27] for the motion control of a hydraulic rotary actuator, a novel model reference adaptive control with the external filter is proposed in [28] to stabilize the system and also guarantee its transient performance, two time-scale filtering technique is applied in [29] to compensate the disturbance for the carbot's moving process.

The contribution of this paper are mainly twofold. First, a novel parallel actuation dual-stage scheme is introduced in this work as mentioned above. Second, an analytical current-force model is established for the coarse actuator Halbach MMLM, which considers the higher order harmonic magnetic field, and based on the proposed model, the force ripple is quantitatively analyzed for the coarse actuator both in theory and in practical. The remaining of this paper is organized as follows. Section II presents the configuration of the proposed parallel actuation dual-stage positioning system, and the modeling of the coarse actuator Halbach MMLM with its analysis of force ripple is detailed in Section III. Section IV describes the control strategy for the proposed dual-stage positioning system, and the experimental results are discussed in Section V.

II. DUAL-STAGE POSITIONING SYSTEM WITH PARALLEL ACTUATION

The schematics of the proposed parallel actuation dual-stage positioning system is shown in Fig. 2(a). The coarse actua-

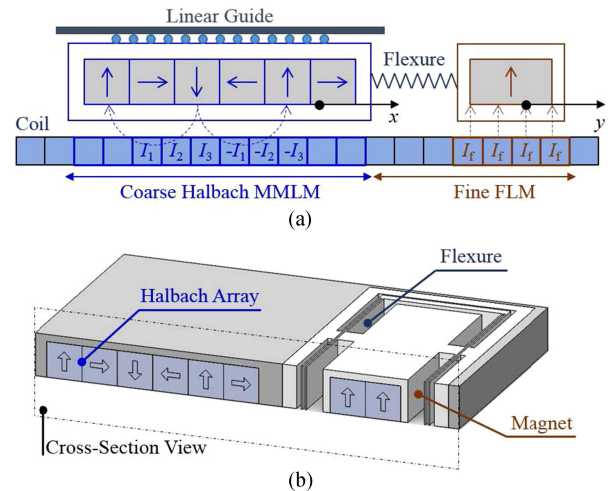


Fig. 2. (a) Schematics of the proposed dual-stage positioning system and (b) conceptual illustration of the translator in the proposed dual-stage system.

tor employs a Halbach MMLM to perform large-displacement coarse motion. This Halbach MMLM consists of two parts, a moving magnet translator and a coil stator. The translator is formed by a Halbach PM array delivering magnetic field and the magnetization directions in Halbach PM array are indicated in Fig. 2(a). Using such arrangement, sinusoidal magnetic field along x - and z - directions are generated and concentrated in the area below the Halbach PM array. The translator of the coarse actuator is supported by the linear guide bearings. The stator of the coarse actuator that lies under the Halbach PM array is a three-phase coil array. By controlling the current energized in the coils, the propulsion force is generated on the translator and leads to a motion. Compared with using the traditional North–South magnet array linear motors as the coarse actuator, the Halbach MMLM is free of cogging force and with better linearity, which will be detailed in Section III.

As the nonlinear effects such as the mechanical friction [30] and commutation error in the Halbach MMLM are still unavoidable, the positioning performance of the coarse motion is limited. In order to enhance the dynamic tracking accuracy of the positioning system, a direct-drive FLM is utilized as the fine actuator to deliver the fine positioning and its schematic is shown in Fig. 2(a). A specific flexure joint, as illustrated in Fig. 2(b), is used to provide frictionless bearing for this fine motion, and the high stiffness characteristic of flexure joint also improve the positioning resolution and bandwidth. The fine actuator also includes a translator and coil stator. The translator is made of PM and supported by the flexure bearing. The coil stator of the fine actuator can be implemented separately with the coarse coil stator. In this case, the fine actuator is only capable to deliver fine positioning within the region of the fine coil stator. Such design is especially suitable for some applications where the positioning system needs to convey the samples over a large-displacement range, but only need to perform the fine positioning within a relatively small working range. On the other hand, the fine actuator can also utilize the same coil sta-

tor as the coarse actuator as shown in Fig. 2, by individually controlling each coil in the stator using some coil switching techniques [31], [32]. In that case, the proposed dual-stage system can deliver both the coarse and fine positioning over an unlimited displacement in theory.

III. ANALYSIS OF FORCE RIPPLE IN COARSE HALBACH MMLM

Force ripple and cogging force are two of the most significant problems that limit the performance of the large-stroke electromagnetic linear actuators. Since the Halbach MMLM is free of iron, the cogging force is ideally avoided from the mechanical design. The reason of the force ripple in the electromagnetic linear actuators is due to the nonsinusoidal magnetic field of their PM arrays, which is caused by the higher order harmonic magnetic fields. In this section, a novel current–force modeling about the coarse Halbach MMLM is established considering the higher order harmonic magnetic field, so that it can be further utilized to quantitatively analyze the force ripple of coarse Halbach MMLM.

A. Force Modeling of Halbach MMLM

In order to establish the relationship between the current and force for the Halbach MMLM, the magnetic field of the Halbach PM array is derived considering the higher order harmonics. The entire space of the Halbach PM array, as shown in Fig. 3, is divided into three regions, namely, Region I ($z \geq h_m$), Region II ($0 \leq z < h_m$), and Region III ($z < 0$). Assuming that the Halbach PM array extends infinitely in the x -axis with infinite number of magnet segments, the magnetization of the Halbach PM array $\mathbf{M}_{II}$ can be expressed in Fourier series that $\mathbf{M}_{II} = [\sum_{n=1}^{+\infty} M_{xn} \sin(\gamma_n x) \quad \sum_{n=1}^{+\infty} M_{zn} \cos(\gamma_n x)]^T$, where γ_n is spatial wave number equal to $2\pi n/L$ and L represents the pitch of the Halbach PM array as indicated in Fig. 3, and $n = 1, 2, 3, \dots$, which denotes the n th harmonic of the Fourier series. Note that the Halbach PM array considered in this modeling is in general form, whereby the width of horizontally magnetized magnet segments can differ from the vertically magnetized segments, here λ is introduced to represent their width difference.

By introducing the magnetic scalar potential φ [33], [34], the governing equations for three regions are $\nabla^2 \varphi_k = \nabla \cdot \mathbf{M}_k$, where k denotes each region and $k = \text{I, II, and III}$, and $\mathbf{M}_I = \mathbf{M}_{III} = 0$. Considering the properties $\lim_{z \rightarrow +\infty} \varphi_I = \lim_{z \rightarrow -\infty} \varphi_{III} = 0$, the general solutions for φ_I , φ_{II} , and φ_{III} are given in the forms

$$\begin{aligned} \varphi_I &= \sum_{n=1}^{+\infty} K_1 e^{-\gamma_n z} \cos(\gamma_n x), \quad \varphi_{III} = \sum_{n=1}^{+\infty} K_4 e^{\gamma_n z} \cos(\gamma_n x), \\ \varphi_{II} &= \sum_{n=1}^{+\infty} \left(K_2 e^{-\gamma_n z} + K_3 e^{\gamma_n z} - \frac{M_{xn}}{\gamma_n} \right) \cos(\gamma_n x). \end{aligned} \quad (1)$$

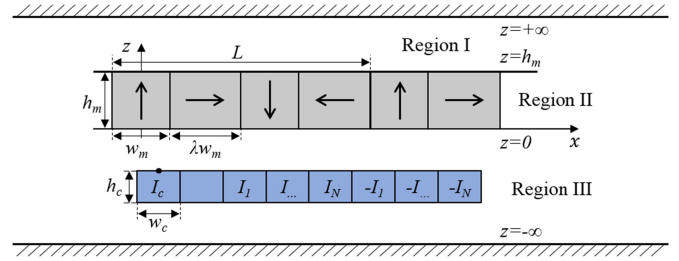


Fig. 3. Three regions for modeling the magnetic field of Halbach PM array with coil array.

The coefficients K_1 , K_2 , K_3 , and K_4 are determined by applying the following boundary conditions:

$$\begin{aligned} H_{Ix}(x, z)|_{z=h_m} &= H_{IIx}(x, z)|_{z=h_m}, \\ B_{Iz}(x, z)|_{z=h_m} &= B_{IIz}(x, z)|_{z=h_m}, \\ H_{IIx}(x, z)|_{z=0} &= H_{IIIx}(x, z)|_{z=0}, \\ B_{IIz}(x, z)|_{z=0} &= B_{IIIz}(x, z)|_{z=0} \end{aligned} \quad (2)$$

where H_{kx} , H_{kz} , B_{kx} , and B_{kz} represent the scalar components of the magnetic field intensity $\mathbf{H}_k$ and magnetic flux density $\mathbf{B}_k$ in x - and z - directions, respectively. $\mathbf{H}_k = -\nabla \varphi_k$, $\mathbf{B}_k = \mu_0(\mathbf{H}_k + \mathbf{M}_k)$, and μ_0 denotes the permeability of the free space.

By substituting (1) into (2), the coefficients can be obtained. The magnetic field of Halbach PM array $\mathbf{B}$ in the interested Region III can be expressed as

$$\begin{aligned} B_{III}(x, z) &= \mu_0 \sum_{n=1}^{+\infty} \frac{(M_{xn} + M_{zn})(1 - e^{-\gamma_n h_m})e^{\gamma_n z}}{2} \\ &\times \begin{bmatrix} \sin(\gamma_n x) \\ -\cos(\gamma_n x) \end{bmatrix} \end{aligned} \quad (3)$$

where the corresponding (x, z) coordinate is indicated in Fig. 3, and the range of x is within the area of the Halbach PM array and $z \leq 0$.

Using the derived magnetic field of the Halbach PM array (3), the magnetic force generated by the coils underneath the array can be analytically derived using the Lorentz-force principle. Assuming one coil with a current magnitude of I_c is located in (x, z) , as shown in Fig. 3, the magnetic force generated by this single coil is calculated as

$$\begin{aligned} \begin{bmatrix} F_x \\ F_z \end{bmatrix} &= \frac{\mu_0 N_t l_m L^2}{4w_c h_c \pi^2} \sum_{n=1}^{+\infty} \\ &\times \left(\frac{e^{\gamma_n z} (M_{xn} + M_{zn})(1 - e^{-\gamma_n h_m})(1 - e^{-\gamma_n h_c})}{n^2} \right. \\ &\left. \sin\left(\frac{\gamma_n w_c}{2}\right) \begin{bmatrix} \cos(\gamma_n x) \\ \sin(\gamma_n x) \end{bmatrix} \right) I_c = \begin{bmatrix} \phi_x(x, z) \\ \phi_z(x, z) \end{bmatrix} I_c \end{aligned} \quad (4)$$

where N_t represents the number of wire turns in single coil, l_m denotes the effective length of the coil, w_c and h_c represent the width and height of a single coil, respectively.

By employing the force model derived for a single coil from (4), the force generated by the coils array in Fig. 3 can be determined. Assuming an N -phase configuration is implemented, the total width of $2N$ coils will equal to the pitch L of the Halbach PM array. The total force generated on the Halbach PM array is expressed as

$$\begin{bmatrix} F_{xt} \\ F_{zt} \end{bmatrix} = -2N_m \begin{bmatrix} \phi_x(x, z) & \phi_x(x + \frac{L}{2N}, z) & \cdots \\ \phi_z(x, z) & \phi_z(x + \frac{L}{2N}, z) & \cdots \end{bmatrix} \begin{bmatrix} I_1 \\ \vdots \\ I_N \end{bmatrix} = \Phi_N(x, z) \mathbf{I}_N \quad (5)$$

where I_j , $j = 1, 2, \dots, N$, corresponds to the current magnitude of the j th phase, $\Phi_N(x, z)$ denotes the 2 by N matrix in (5), $\mathbf{I}_N = [I_1 \ I_2 \ \dots \ I_N]^T$ represents the current vector, and N_m is the number of the Halbach PM array period.

B. Sinusoidal Commutation of Halbach MMLM

For the Halbach MMLM implemented in this paper, the case $N = 3$ and $\lambda = 1$ is adopted. Hence, the force generated for a single coil (4) is simplified to

$$\begin{bmatrix} F_x \\ F_z \end{bmatrix} = \frac{\sqrt{2}M_0\mu_0 l_m L^2 N_t}{w_c h_c \pi^3} \sum_{i=0}^{\infty} (-1)^{i+1} \left(\frac{e^{\gamma_n z}}{n^3} (1 - e^{-\gamma_n h_m}) \right) (1 - e^{-\gamma_n h_c}) \sin\left(\frac{\gamma_n w_c}{2}\right) \begin{bmatrix} \cos(\gamma_n x) \\ \sin(\gamma_n x) \end{bmatrix} I_c. \quad (6)$$

The sinusoidal current commutation is derived based on the force model (6) by neglecting the higher order harmonics, i.e.

$$\begin{aligned} \mathbf{F} &= \kappa \begin{bmatrix} \cos(\gamma_1 x) & \cos(\gamma_1 x - \pi/3) & \cos(\gamma_1 x - 2\pi/3) \\ \sin(\gamma_1 x) & \sin(\gamma_1 x - \pi/3) & \sin(\gamma_1 x - 2\pi/3) \end{bmatrix} \mathbf{I}_3 \\ &= \tilde{\Phi}_3(x) \mathbf{I}_3 \end{aligned} \quad (7)$$

where $\mathbf{F}$ denotes the total force vector $[F_{xt} \ F_{zt}]^T$, $\tilde{\Phi}_3$ is the mapping matrix neglecting higher order harmonics, which differs from Φ_3 in (5), and

$$\kappa = -\frac{2\sqrt{2}M_0\mu_0 l_m L^2 N_t N_m (1 - e^{-\gamma_1 h_m})(1 - e^{-\gamma_1 h_c}) e^{\gamma_1 z_0}}{w_c h_c \pi^3} \quad (8)$$

is a constant gain obtained from (4) with detailed specifications, and z_0 is fixed in this work as the moving magnet array is supported by linear guides.

Consequently, the sinusoidal current commutation law is given as

$$\mathbf{I}_3 = \tilde{\Phi}_3^T(x) (\tilde{\Phi}_3(x) \tilde{\Phi}_3^T(x))^{-1} \mathbf{f} = \tilde{\Psi}_3(x) \mathbf{f} \quad (9)$$

where $\mathbf{f}$ represents the desired force that $\mathbf{f} = [f_{xt} \ f_{zt}]^T$, $\tilde{\Psi}_3(x) = \tilde{\Phi}_3^T(x) (\tilde{\Phi}_3(x) \tilde{\Phi}_3^T(x))^{-1}$, and it can be directly ob-

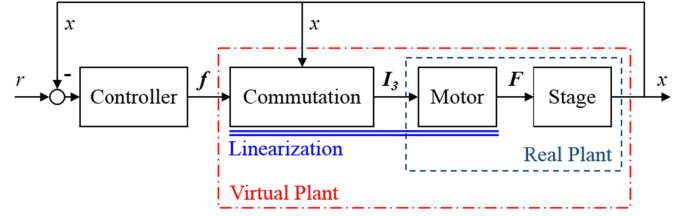


Fig. 4. Control structure of the Halbach MMLM system.

tained as

$$\tilde{\Psi}_3(x) = \frac{2}{3\kappa} \begin{bmatrix} \cos(\gamma_1 x) & \sin(\gamma_1 x) \\ \cos(\gamma_1 x - \pi/3) & \sin(\gamma_1 x - \pi/3) \\ \cos(\gamma_1 x - 2\pi/3) & \sin(\gamma_1 x - 2\pi/3) \end{bmatrix}. \quad (10)$$

Overall, the control structure of the Halbach MMLM is illustrated as in Fig. 4. The real physical input is the three-phase current $\mathbf{I}_3$ obtained from the commutation law which linearizes the Halbach MMLM. The input current $\mathbf{I}_3$ generates the force $\mathbf{F}$ and finally leads to the output displacement x .

C. Case Study: Model-Based Analysis of Force Ripple

The proposed force model (6) is capable to analyze the force ripple of the MMLM with the general Halbach PM array. In this case study, the analysis of the force ripple is carried out for the standard Halbach PM array with $\lambda = 1$ in Fig. 3, while for other configuration of the Halbach PM array, the analysis can be performed in a similar way with different Fourier coefficients.

According to the above modeling, it is noted that the theoretical reason for the force ripple is due to the linearization error from neglecting the higher order harmonic magnetic field in (5), where the $\tilde{\Phi}_3^T(x)$ in the sinusoidal commutation law $\tilde{\Psi}_3(x) = \tilde{\Phi}_3^T(x) (\tilde{\Phi}_3(x) \tilde{\Phi}_3^T(x))^{-1}$ is different from the real $\Phi_3(x)$, so that $\mathbf{F} = \tilde{\Phi}_3(x) \mathbf{I}_3 = \Phi_3(x) \tilde{\Psi}_3(x) \mathbf{f} \neq \mathbf{f}$. As a result, the position-dependant periodical error between the real force $\mathbf{F}$ and the desired force $\mathbf{f}$ causes the force ripple phenomenon.

To minimize such force ripple, one method is to design the better Halbach PM arrays to reduce the magnitude of higher order harmonics, e.g., using eight-segment Halbach PM array instead of four-segment one [35], [36]. However, such design increases both the manufacturing cost and assembly complexity. On the other hand, the commutation law can also be replaced by $\Psi_3(x) = \Phi_3^T(x) (\Phi_3(x) \Phi_3^T(x))^{-1}$ to include the higher order harmonics. However, the nonsinusoidal commutation is not preferred in practical as it involves the real-time matrix inversion.

The modeling approach presented in this work is capable to quantitatively analyze the higher order harmonics field and the resulted force ripple due to this neglecting, as (6) models all harmonic force components. Fig. 5(a) shows the comparison between the fundamental harmonic force and the force containing higher order harmonics in the adopted motor design in this paper with 0.5 mm air gap, which indicates that the force errors due to the negligence of higher order harmonics is less than 1.5%. This

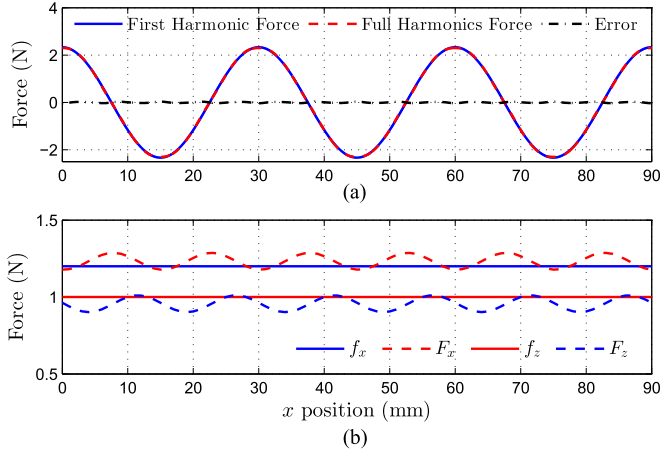


Fig. 5. (a) Comparison between the first harmonic force and force containing higher order harmonics, (b) comparison between the desired force $[f_{tx} \ f_{tz}]^T$ with the real generated force $[F_{tx} \ F_{tz}]^T$ in the presence of a small misalignment.

small difference can also be explained by the $1/n^3$ term in each harmonic force component of (6), where the following nonzero components in the standard Halbach PM array are $n = 5, 9, \dots$. The above analysis indicates that the theoretical force ripple due to the neglecting of higher order harmonics is small, and the Halbach MMLM using sinusoidal current commutation is almost free of force ripple even using the four-segment Halbach PM array in theory.

However, in practical, as both the coil array and Halbach PM array utilized in this work are fabricated by stacking the coils and PMs one after another, the assembly misalignment is unavoidable in manufacturing. Such misalignment will change the ideal current-force relationship (7), which also results the force ripple during linearization. To illustrate this effect, Fig. 5(b) shows the simulation of generating constant f , while (7) is affected by a small misalignment, i.e., $\Delta = 0.5$ mm in the second column of $\tilde{\Phi}_3(x)$. Compared with Fig. 5(a), this indicates that the force ripple due to the effects of assembly misalignment might be more significant than the effect of neglecting higher order harmonics. This analysis suggests that the precise fabrication, assembly, and calibration of the Halbach MMLM is important to avoid the force ripple, especially for identifying the initial home position.

IV. CONTROL STRATEGY FOR DUAL-STAGE POSITIONING

In this section, the control strategy for the dual-stage positioning is presented. As the feedback signals are available both from the coarse and fine actuators in this work, the dual-feedback control scheme, as shown in Fig. 6, is adopted to control the dual-stage positioning system.

A. Controller Design for Coarse Motion

Although the coarse actuator suffers less from the force ripple and cogging force as presented, some other disturbance, e.g., the mechanical friction, is still unavoidable.

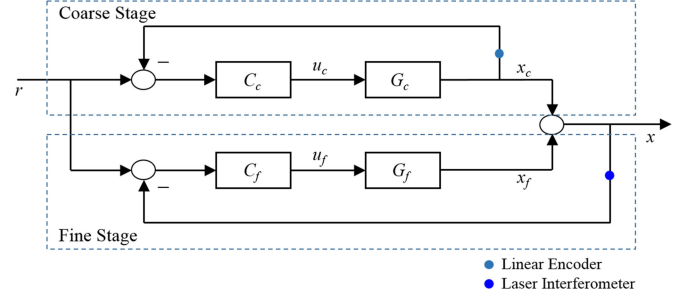


Fig. 6. Dual feedback configuration for the dual-stage positioning system.

In this work, the controller for the coarse motion is proposed by combining the nominal PID controller with a disturbance observer and its asymptotical stability is ensured by an additional sliding mode term. Consider the possible linearization error as discussed, the dynamic equation of coarse actuator can be formulated as

$$m_c \ddot{x} + d_c \dot{x} + k_c x = (g_c + \Delta g_c) u_c + f \quad (11)$$

where Δg_c represents the additive uncertainty in g_c , and f represents all other disturbances, such as the mechanical friction, coupling, etc. Define the state vector $x = [x_1 \ x_2 \ x_3]^T$ and (11) is given in the state-space form as

$$\dot{x} = Ax + Bu_c + Bd \quad (12)$$

where A and B can be obtained from (11) directly and $d = (\Delta g_c u_c + f)/g_c$.

The following disturbance observer [37] is utilized to estimate the disturbance d in the coarse motion

$$\begin{aligned} \dot{p} &= -LB(p + Lx) - L(Ax + Bu_c) \\ \hat{d} &= p + Lx \end{aligned} \quad (13)$$

where $\hat{d}$ denotes the estimation of the disturbance, p is the internal state of the disturbance observer, and L is the gain of the observer.

To facilitate the coarse actuator to track a desired signal x_r , the state-space model (12) is expressed in an error equation form

$$\dot{\eta} = A\eta + Bu_c - Bf_r + Bd \quad (14)$$

where $\eta = [\eta_1 \ \eta_2 \ \eta_3]^T$, $\eta_1 = x_1 - \int_0^t x_r dt$, $\eta_2 = x_2 - x_r$, $\eta_3 = x_3 - \dot{x}_r$, and $f_r = \ddot{x}_r + a_1 \dot{x}_r + a_2 x_r$. Here, the reference x_r is assumed to be twice differentiable.

Based on the estimation of disturbance, the following control law is used for the coarse motion:

$$u_c = K\eta + f_r - \hat{d} - k_s \text{sign}(\eta^T B) \quad (15)$$

where $\text{sign}(\cdot)$ denotes the signum function, $K = [k_i \ k_p \ k_d]$ is the vector of the nominal PID controller parameters that ensure $A + BK < 0$, and $-k_s \text{sign}(\eta^T B)$ is the sliding mode term to ensure the closed-loop stability in the presence of the disturbance estimation error.

To analyze the stability of the closed-loop system, the following assumptions are made:

Assumption 1: The derivative of the disturbance in system (12) is bounded and satisfies $|\dot{d}| < \epsilon$.

Assumption 2: Define the estimation error of disturbance observer as $\tilde{d} = d - \hat{d}$ and let $\tilde{d}_u$ be the upper bound of all possible absolute initial estimation error that $\tilde{d}_u = \max(|\tilde{d}(t_0)|)$, it is assumed that $\frac{\epsilon}{LB} < \tilde{d}_u$ is valid.

Theorem 1: Consider the dynamic system (14) with the disturbance observer (13) starting at time t_0 , the estimation error $\tilde{d}$ will finally be ultimately bounded that $|\tilde{d}| < \bar{k}_s$ for all time $t \geq t_0 + \frac{1}{LB} \ln(\frac{LB\tilde{d}_u - \epsilon}{LB\bar{k}_s - \epsilon})$ by choosing the observer gain L , such that $LB > 0$ holds, where $\bar{k}_s$ meets the condition $\frac{\epsilon}{LB} < \bar{k}_s < \tilde{d}_u$. Furthermore, by applying the control law (15) after $t \geq t_0 + \frac{1}{LB} \ln(\frac{LB\tilde{d}_u - \epsilon}{LB\bar{k}_s - \epsilon})$, the closed-loop system is asymptotically stable if the switching gain k_s in (15) is selected, such that $k_s > \bar{k}_s$.

Proof: To prove the ultimate boundedness of the estimation error for the disturbance observer, a candidate Lyapunov function is defined as $V_d = \tilde{d}^2/2$. Take the derivative of V_d

$$\begin{aligned}\dot{V}_d &= \tilde{d}(\dot{d} - \dot{\hat{d}}) \\ &= \tilde{d}(\dot{d} - \dot{p} - L\dot{x}) \\ &= \tilde{d}(\dot{d} + LB(p + Lx) + L(Ax + Bu_c) \\ &\quad - L(Ax + Bu_c + Bd)) \\ &= -LB\tilde{d}^2 + \dot{d}\tilde{d}.\end{aligned}\quad (16)$$

As $\dot{d}$ is bounded that $|\dot{d}| < \epsilon$, the foregoing equality (16) implies that

$$\dot{V}_d \leq -(1 - \theta)LB\tilde{d}^2, \quad \forall |\tilde{d}| > \frac{\epsilon}{\theta LB} \quad (17)$$

where $0 < \theta < 1$. According to $LB > 0$, the equality (17) shows that the estimation error of disturbance observer $\tilde{d}$ will reach the set

$$\Omega_d = \left\{ V_d(\tilde{d}) \leq \frac{\epsilon^2}{2\theta^2 \|LB\|_2^4} \right\} \quad (18)$$

in finite time. Furthermore, according to the comparison lemma [38], $\tilde{d}(t)$ satisfies the bound

$$\begin{aligned}|\tilde{d}(t)| &\leq \tilde{d}_u e^{-LB(t-t_0)} + \epsilon \int_{t_0}^t e^{-LB(t-\tau)} d\tau \\ &= \tilde{d}_u e^{-LB(t-t_0)} + \frac{\epsilon}{LB} (1 - e^{-LB(t-t_0)}).\end{aligned}\quad (19)$$

As a result, for any $\bar{k}_s$ picked that satisfies $\frac{\epsilon}{LB} < \bar{k}_s < \tilde{d}_u$

$$|\tilde{d}(t)| \leq \bar{k}_s, \quad \forall t \geq t_0 + \frac{1}{LB} \ln \left(\frac{LB\tilde{d}_u - \epsilon}{LB\bar{k}_s - \epsilon} \right). \quad (20)$$

Next, $V_\eta = \eta^T \eta/2$ is defined as another Lyapunov function, take the derivative of V_η for any $t \geq t_0 + \frac{1}{LB} \ln \left(\frac{LB\tilde{d}_u - \epsilon}{LB\bar{k}_s - \epsilon} \right)$

$$\begin{aligned}\dot{V}_\eta &= \eta^T \dot{\eta} \\ &= \eta^T (A\eta + B(K\eta + f_r - \hat{d} - k_s \text{sign}(\eta^T B)) - Bf_r + Bd) \\ &= \eta^T (A\eta + BK\eta + B(d - \hat{d}) - k_s B \text{sign}(\eta^T B)) \\ &= \eta^T (A + BK)\eta + \eta^T B\tilde{d} - k_s \eta^T B \text{sign}(\eta^T B) \\ &\leq \eta^T (A + BK)\eta + (\bar{k}_s - k_s) |\eta^T B|.\end{aligned}\quad (21)$$

Since $\eta^T (A + BK)\eta < 0$ and $k_s > \bar{k}_s$, it leads to

$$\dot{V}_\eta < 0 \quad (22)$$

which implies that the impact caused by the disturbance estimator error $\tilde{d}$ can be rejected by the sliding mode term $-k_s \text{sign}(\eta^T B)$ and the tracking error of the closed-loop system will finally converge to zero. This completes the proof. ■

Remark 1: To further eliminate the chattering phenomenon caused by the discontinuity of the signum function $\text{sign}(\eta^T B)$, the signum function in (15) can be replaced with the saturation function $\text{sat}(\frac{\eta^T B}{\epsilon})$, where

$$\text{sat} \left(\frac{\eta^T B}{\epsilon} \right) = \begin{cases} \frac{\eta^T B}{\epsilon} & \text{if } \left| \frac{\eta^T B}{\epsilon} \right| \leq 1 \\ \text{sgn} \left(\frac{\eta^T B}{\epsilon} \right) & \text{if } \left| \frac{\eta^T B}{\epsilon} \right| > 1 \end{cases} \quad (23)$$

where ϵ is a positive constant. In this case, when $|\eta^T B| > \epsilon$, (21) remains unchanged and $\dot{V}_\eta < 0$, which indicates that $|\eta^T B|$ will be strictly decreasing, until it reaches the set $\{|\eta^T B| \leq \epsilon\}$ and remains inside thereafter. Then, (21) becomes

$$\begin{aligned}\dot{V}_\eta &= \eta^T (A + BK)\eta + \eta^T B\tilde{d} - k_s (\eta^T B)^2 / \epsilon \\ &\leq \eta^T (A + BK)\eta + \bar{k}_s |\eta^T B| - k_s (\eta^T B)^2 / \epsilon \\ &\leq \eta^T (A + BK)\eta - (1 - \nu) k_s (\eta^T B)^2 / \epsilon, \\ &\quad \forall |\eta^T B| > \frac{\bar{k}_s \epsilon}{\nu k_s}\end{aligned}\quad (24)$$

where $0 < \nu < 1$, and this indicates that the system will achieve ultimate boundedness within the boundary of set $\{|\eta^T B| \leq \frac{\bar{k}_s \epsilon}{\nu k_s}\}$.

B. Closed-Loop System of Fine Motion

The mechanical plant of fine actuator and its nominal controller are designed by applying an integrated servo-mechanical design approach [13]. Without loss of generality, the plant of fine actuator is modeled as

$$G_f(s) = \frac{g_f}{m_f s^2 + d_f s + k_f} = \frac{n_p}{d_p(\lambda_p)} \quad (25)$$

where d_f and k_f are the plant design parameters to be determined, n_p and d_p denote the plant numerator and denominator polynomials, respectively, and λ_p denotes the unknown design

parameters in the plant. A standard PID controller is utilized for the fine stage with the structure

$$C_f(s) = K_p + \frac{K_i}{s} + \frac{K_d s}{1 + T_d s} = \frac{n_k(\lambda_k)}{d_k} \quad (26)$$

where K_p , K_i , and K_d are the PID controller parameters to be determined, and T_d is a fixed known constant. n_k and d_k are the controller numerator and denominator polynomials, respectively, and λ_k denotes the unknown controller parameters. The integrated servo-mechanical design approach is performed on the objective to achieve desired closed-loop specifications, by carrying out a convex optimization. The optimization is formulated as

$$\begin{aligned} & \min_{(\lambda_p \text{ and } \lambda_k)} \rho_S \\ & \text{subject to} \\ & \left\{ \begin{array}{l} \frac{n_p n_k(\lambda_k) + d_p(\lambda_p) d_k}{d_c} \in \mathbb{S}, \\ |S_f(j\omega)| < \rho_S \quad \omega \in \Omega_S, \\ |T_f(j\omega)| < \rho_T \quad \omega \in (\omega_h, +\infty), \\ \frac{g_f i_r}{x_r} < k_f < \frac{\bar{F}_c}{x_s}, \\ \underline{d}_f < d_f < \bar{d}_f. \end{array} \right. \end{aligned} \quad (27)$$

The first constraint in (27) ensures the stability of the closed-loop fine system, where d_c is a stable central polynomial and $\mathbb{S}$ denotes the set of all proper positive-real transfer functions. The second and third constraints in (27) denote the disturbance rejection specification and high-frequency roll-off specification, respectively, where S_f and T_f represent the sensitivity and complementary sensitivity transfer functions of the fine stage, respectively. ρ_S is the performance index to be minimized and ρ_T is a constant value set as -3 dB, and Ω_S denotes the target frequency range of disturbance rejection. The fourth constraint specifies the feasible range of k_f , which is determined from the consideration of both the open-loop positioning resolution x_r and the maximal coarse-fine coupling force $\bar{F}_c$, the rest symbols in this constraint x_s and i_r denote the displacement stroke of fine actuator and the resolution of current amplifier, respectively. Finally, the fifth constraint specifies the feasible range of d_f that ensures the obtained d_f can be realized by selection of flexure material or using additional eddy current damper.

The basic motivation of this design approach is to choose the primary resonant mode of the fine actuator (by adjusting the stiffness of flexure) to locate at such sensitivity peak frequency and then phase-stabilize this resonant mode. Consequently, the designed fine motion can provide strong disturbance attenuation ability for such specific disturbance, which is caused by the sensitivity peak of coarse motion. Furthermore, the integrated design approach also ensures other control specifications such as stability and high-frequency roll-off conditions and some mechanical specifications such as open-loop positioning resolution and maximal acceleration as formulated in (27).

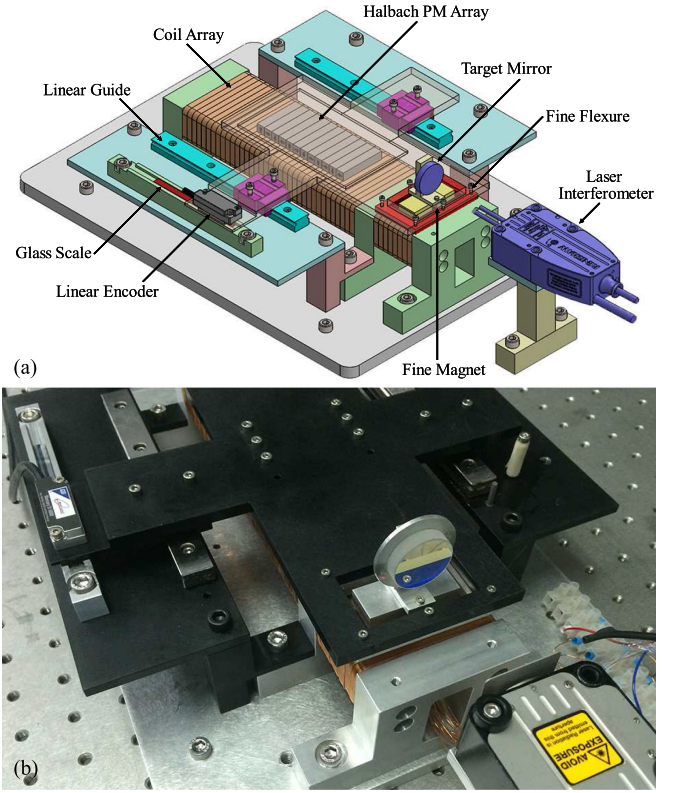


Fig. 7. (a) Structure layout and (b) photo of the dual-stage prototype.

TABLE I
MECHANICAL PARAMETERS IN PROTOTYPE

Height of Rectangular Coil, h_c	4 mm
Width of Rectangular Coil, w_c	5 mm
Turn Numbers of Rectangular Coil, N_t	100
Period Numbers of Halbach Array, N_m	3
Width of PMs in Halbach Array, w_m	7.5 mm
Length of PMs in Halbach Array, l_m	40 mm
Magnetization Magnitude, M_0	1.3 T

V. EXPERIMENTS AND DISCUSSIONS

A prototype is fabricated to evaluate the feasibility and performance of the proposed dual-stage positioning system, and the design target is to investigate how is the improvement by using the proposed dual-stage design compared with the coarse stage and whether the prototype can reach 50 nm positioning accuracy. The structure layout of the dual-stage prototype is depicted in Fig. 7(a), and its photo is shown in Fig. 7(b). The coarse Halbach MMLM is mechanically supported by two parallel linear guides, and its displacement is measured by a MicroE Mercury II 5000 linear encoder with 20 nm resolution. The fine stage is installed on the coarse translator via the specially designed flexure, and its motion is sensed by a Renishaw RLE fibre optic laser interferometer with a resolution of 10 nm. The key mechanical parameters of the prototype are listed in the Table I. The current dual-stage prototype enables a 10 mm motion stroke for evaluation, which can be easily extended by using more coils or adopting the coil switching technique. A National Instruments

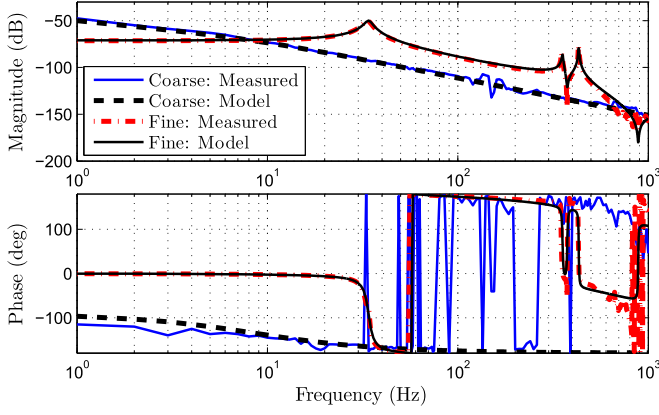


Fig. 8. Measured frequency response of the coarse and fine actuators.

TABLE II
IDENTIFIED PARAMETERS OF FINE ACTUATOR

n	k_n	ζ_n	ω_n (rad/s)
1	2.82×10^{-4}	4.41×10^{-2}	$2\pi \ 34.02$
2	-7.20×10^{-7}	7.50×10^{-3}	$2\pi \ 354.80$
3	-8.96×10^{-7}	4.00×10^{-3}	$2\pi \ 431.50$

PXI-8110 platform is utilized as the controller with a 10 kHz sampling rate and Trust Automation TA320 and TA115 linear current amplifiers are employed to actuate the coarse and fine actuators, respectively. For the current prototype, the actuation force for coarse motion can be more than 25 N, and the achievable acceleration can be larger than 50 m/s².

Fig. 8 shows the measured frequency response of both the coarse and fine actuators in frequency range of 1–1000 Hz. The coarse actuator's modal parameters in (11) are identified as $m_c = 1$, $d_c = 54$, $k_c = 0$, and $g_c = 1.08$. The observer gain L in (13) is set as $[0 \ 0 \ 0.86]^T$, the nominal PID parameters is chosen as $K = \text{diag}([557142 \ 39000 \ 68.3])$, and k_s in (15) is selected as 0.015. The plant model of the fine actuator is also identified as

$$G_f = \left(\frac{2 - T_d s}{2 + T_d s} \right) \sum_{n=1}^3 \frac{k_n \omega_n^2}{s^2 + 2\zeta_n \omega_n s + \omega_n^2} \text{ A/m} \quad (28)$$

where $T_d = 2.25 \times 10^{-4}$ s and the parameters in each mode are listed in Table II. The nominal controller of fine motion is selected as

$$C_f(s) = 2.14 \times 10^3 + \frac{8.01 \times 10^5}{s} + \frac{18.23s}{1 + s/990}. \quad (29)$$

From Fig. 8, it is also observed that, compared with the coarse actuator with zero stiffness, the flexure-based fine actuator is with relatively larger stiffness k_f and less mass m_f , so its primary resonant mode is much higher than the coarse actuator according to $\omega_f = \sqrt{k_f/m_f}$. Furthermore, the flexure-based fine actuator is of better linearity, and as a result, the achievable bandwidth of fine actuator is much higher than the coarse actuator.

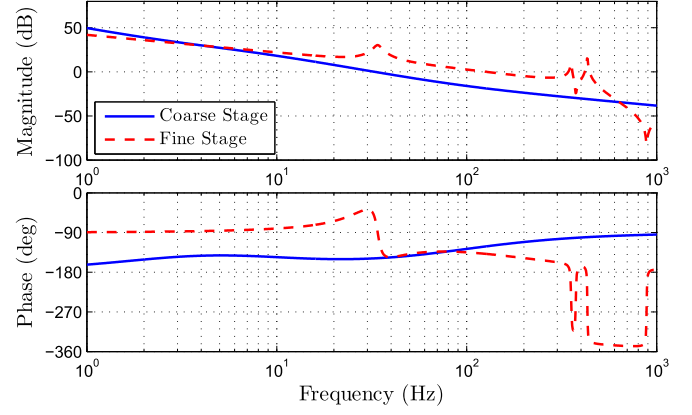


Fig. 9. Open-loop transfer function of the coarse motion (with nominal PID controller) and fine motion.

During the integration of the dual-stage positioning system, it is noted that the nominal open-loop transfer function of fine motion is with large primary and secondary phase margins, which indicates the nominal fine motion possesses strong robustness. For the specific prototype in this paper, to further improve the bandwidth of fine motion, a tuned constant gain, $\alpha_f = 3.5$, is attached to the nominal fine motion controller, so that $\bar{C}_f = \alpha_f C_f$ is implemented for the fine actuator. As the constant gain α_f does not affect the open-loop phase property of fine motion, the stability is still ensured by the integrated design approach as the two high-frequency resonant modes are phase-stabilized. Fig. 9 shows the open-loop transfer functions of the coarse motion (with nominal PID controller) and fine motion. It indicates that the crossover frequency between coarse and fine motion is around 5 Hz and their phase difference is around 60° at the crossover frequency. The open-loop bandwidth is enhanced from around 30 Hz in the coarse motion to around 130 Hz in the fine motion.

The dual-stage positioning system is used to track a 1 mm magnitude sinusoidal reference signal at 1 Hz as shown in Fig. 10(a), the estimated disturbance is plotted in Fig. 10(b) and the control signals of both the coarse and fine motion are shown in Fig. 10(c) and (d), respectively. The positioning error signals of both the coarse- and dual-stage positioning systems are plotted in Fig. 11(a) and (b), respectively. The histograms of the tracking error are also calculated and shown in Fig. 11(c) and (d). Define the root mean square error (RMSE) as $\|e\|_2 = \sqrt{\frac{1}{N} \sum_{i=1}^N e_i^2}$. The RMSE of the coarse and dual-stage systems are calculated, $\|e_c\|_2 = 22.7 \mu\text{m}$ and $\|e_d\|_2 = 0.45 \mu\text{m}$, which indicates that the dual-stage positioning system can significantly reduce the tracking error of the coarse actuator.

The point-to-point positioning performance is also evaluated experimentally and Fig. 12(a) shows the 5mm s-curve reference signal [39]. The positioning error signals of both coarse- and dual-stage positioning systems are plotted in Fig. 12(b) and (c), respectively, which demonstrates that the 6 μm static tracking error of coarse motion can be effectively compensated by the

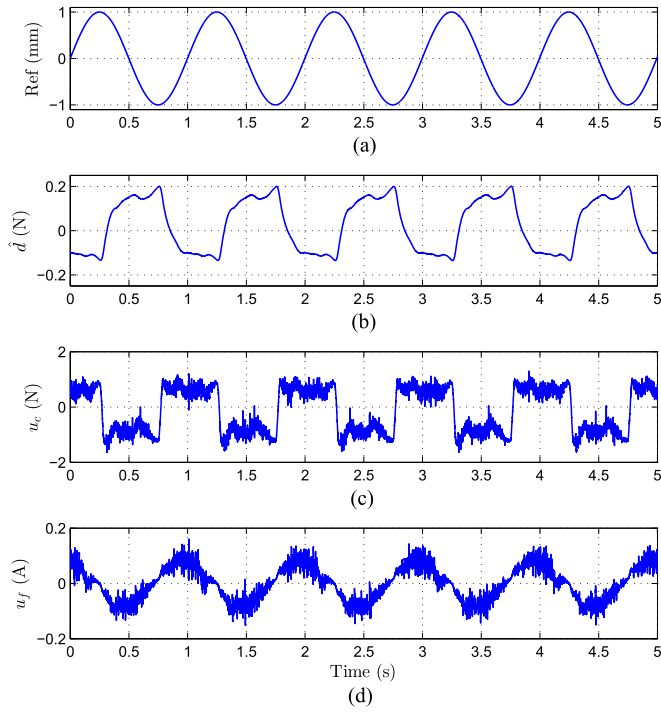


Fig. 10. (a) Reference signal, (b) estimated disturbance, (c) control signal of coarse motion, and (d) control signal of fine motion.

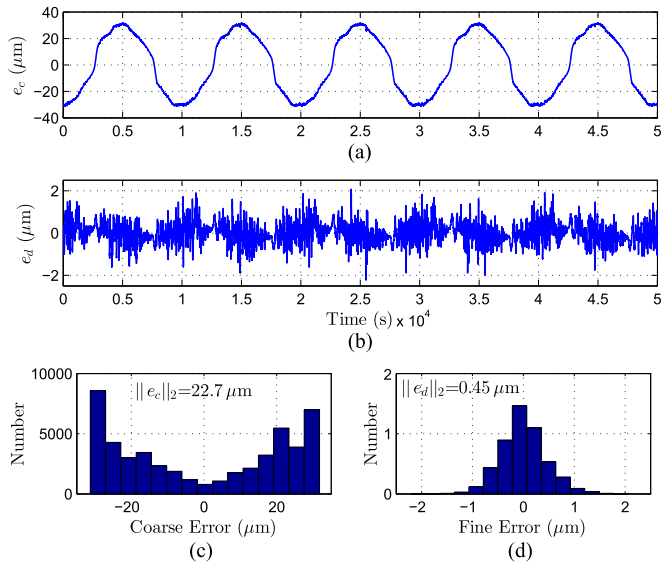


Fig. 11. Tracking error of (a) coarse motion and (b) fine motion, tracking error histograms of the (c) coarse and (d) dual-stage motion.

fine motion. The tracking error of dual-stage positioning system is enlarged in Fig. 12(d) and it shows that the error of point-to-point positioning can achieve less than 40 nm. The positioning resolution of the dual-stage positioning system is also evaluated in Fig. 13 to track a series of 20 nm steps, the result shows that the proposed parallel actuation dual-stage positioning system's resolution can achieve at least 20 nm and the RMSE of the static positioning is 13.15 nm.

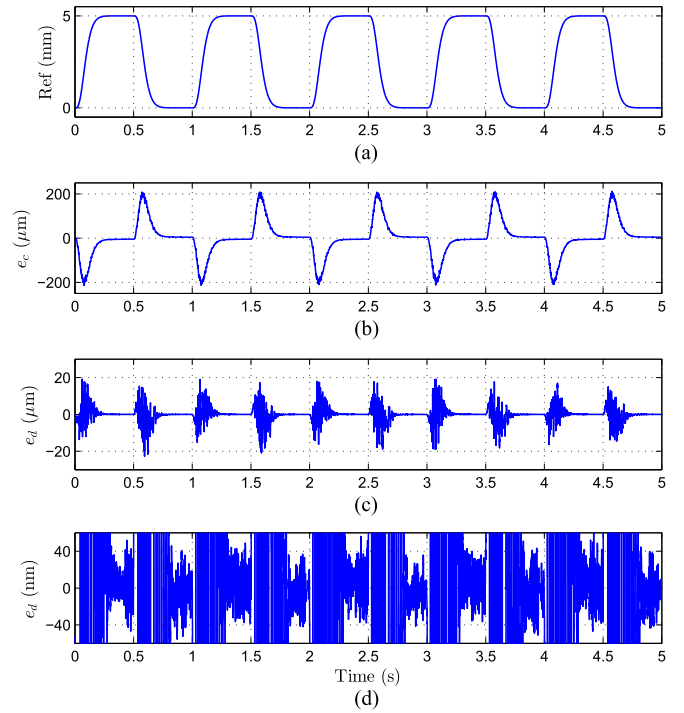


Fig. 12. (a) Reference signal, (b) coarse tracking error, (c) dual-stage tracking error, and (d) enlarged dual-stage tracking error.

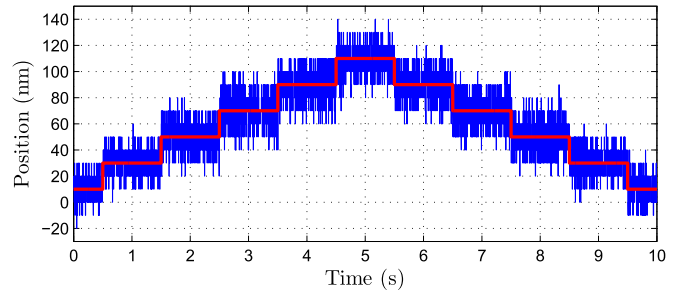


Fig. 13. 20 nm step positioning resolution of the dual-stage system.

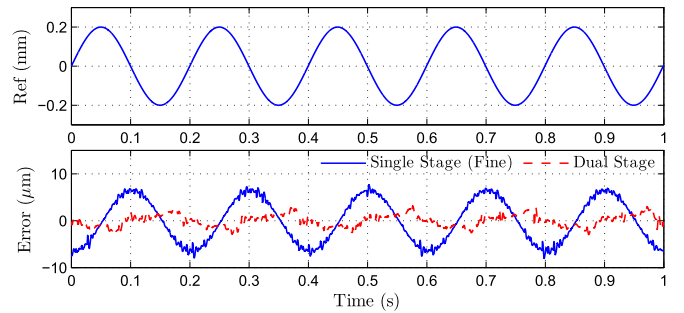


Fig. 14. Comparison of the tracking performance between the single (fine) servo and dual-stage servo.

Finally, Fig. 14 shows the comparison of the tracking performance between the single (fine) servo and dual-stage servo, where a 0.2 mm 5 Hz sinusoidal reference is used. From Fig. 14, it shows that the RMSE of the single (fine) and dual-stage servo are 1.29 and 4.59 μm , respectively. The reason for the im-

provement of dual-stage servo is that the coarse stage provides additional disturbance rejection ability in dual servo, as the sensitivity transfer function of dual-stage system S_d is the product of the sensitivity transfer functions of coarse (S_c) and fine (S_f) stages, i.e., $S_d = S_c S_f$.

VI. CONCLUSION

In this paper, a novel design of dual-stage positioning system with parallel actuation was proposed for large-stroke nanopositioning, and the proposed structure offers a solution of cable-free design for the dual-stage positioning. Furthermore, the force ripple of the coarse Halbach MMLM was quantitatively analyzed based on the proposed analytical current–force model considering higher order harmonic magnetic field. The results showed that the force ripple in the Halbach MMLM is not obvious even by using the standard four-segment Halbach PM array. The dual feedback configuration was utilized to control the dual-stage system, and a disturbance observer was employed in the coarse motion to deal with the disturbance with the proofed stability. The experimental results showed that the proposed dual-stage system is capable to achieve 20 nm step resolution with the RMSE of 13.15 nm, and the 5 mm point-to-point positioning error can achieve less than 40 nm.

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